

Balancing Chemical Equations

Counting atoms, balancing equations and writing equations from words

Name _____ Class _____ Date _____

STEPS FOR BALANCING

1. Write correct formulas. 2. Count atoms on each side.
3. Balance one element at a time, leaving H, O and free elements until last.
4. Re-count everything. 5. Use the smallest whole numbers.

Change coefficients only. Never change subscripts.

A. Understanding

1. Explain why a chemical equation must be balanced.

2. What is the difference between the 2 in $2\text{H}_2\text{O}$ and the 2 in H_2O ? Why can you change only one of them when balancing?

B. Counting atoms

3. Count the atoms of each element in (a) 4NH_3 (b) $3\text{Mg}(\text{OH})_2$ (c) $2\text{Fe}_2(\text{SO}_4)_3$

C. Balancing

4. Balance: (a) $\text{N}_2 + \text{H}_2 \longrightarrow \text{NH}_3$ (b) $\text{Mg} + \text{O}_2 \longrightarrow \text{MgO}$ (c) $\text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{O} + \text{O}_2$ (d) $\text{Fe} + \text{Cl}_2 \longrightarrow \text{FeCl}_3$

5. Balance: (a) $\text{Al} + \text{O}_2 \longrightarrow \text{Al}_2\text{O}_3$ (b) $\text{KClO}_3 \longrightarrow \text{KCl} + \text{O}_2$ (c) $\text{Zn} + \text{HCl} \longrightarrow \text{ZnCl}_2 + \text{H}_2$

6. Balance these combustion reactions: (a) $\text{C}_2\text{H}_4 + \text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O}$ (b) $\text{C}_2\text{H}_5\text{OH} + \text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O}$ (c) $\text{C}_4\text{H}_{10} + \text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O}$

7. Balance, treating polyatomic ions as units: (a) $\text{AgNO}_3 + \text{CaCl}_2 \longrightarrow \text{AgCl} + \text{Ca}(\text{NO}_3)_2$ (b) $\text{H}_2\text{SO}_4 + \text{NaOH} \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$ (c) $\text{Al}_2(\text{SO}_4)_3 + \text{Ca}(\text{OH})_2 \longrightarrow \text{Al}(\text{OH})_3 + \text{CaSO}_4$

D. From words to equations

8. Sodium reacts with water to form sodium hydroxide and hydrogen gas. Write the balanced equation.

E. Challenge

9. (a) Is $\text{CaCO}_3 + \text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$ balanced? If not, balance it. (b) Balance the equation for photosynthesis: $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + \text{O}_2$